Swimming Pools Get Greener

Technology innovations proven to reduce energy consumption and environmental impacts in other industries are now gaining acceptance in the U.S. pool-building industry.

Variable-frequency drive technology, widely adopted in other industries that demand hydraulic efficiency, from dairy farms to water treatment plants, is now being adapted into variable-drive and variable-speed pool pumps. Other equipment, such as electrolytic chlorine systems are experiencing brisk sales.


1.

Pool owners today can reduce their energy usage dramatically and save hundreds of dollars per year by implementing easy and affordable retrofit projects (see "Efficiency Options and Opportunities"). With today's rising energy costs and new state regulations for energy efficiency, everyone involved in the pool industry is wishing to create and environmentally friendly products.

Pumping Out Energy Savings

A Pacific Gas & Electric Company (PG&E) study, conducted by Davis Energy Group and submitted to the California Energy Commission as part of its 2006 Appliance Standards Rulemaking docket determined that the average residential pool pump consumes 2,600 kWh annually. With an estimated 1.5 million private residential in-ground pools in California, it takes the output of six medium-sized power plants just to operate the state's pools.

The PG&E study concluded that pool pumps "are almost always the largest single electrical end-use [appliance in a home]." Data from this study were later integrated into the California Energy Commission's Statewide Appliance Saturation study, along with findings from the Los Angeles Department of Water and Power (LADWP), Southern California Edison (SCE), and San Diego Gas & Electric Company (SDG&E). The results are shown in Table 1.

The state legislature and the state utilities targeted pool systems in the 2005 rewrite of the California Energy Code, making Title 20 and Title 24 common language among the state's pool builders and equipment manufacturers. Those new regulations went into effect this year, placing prescriptive requirements on pool pumps and to a lesser degree, on heaters. This prohibited the sale and installation of some of the most common types of pumps and motors, requiring any 1 hp or larger pump to have two speed settings or variable-speed drives. It also established a regulatory and certification system backed by fines and criminal charges.

The pool experts at PG&E have been getting a lot of attention in New York, New Jersey, Texas, Arizona, and Florida—all of which are considering adopting similar legislation. PG&E and the Florida Swimming Pool Association contributed language to the recently signed Florida Energy Bill (House Bill 7135). This act is nearly identical to the California legislation, although it is slightly lighter on enforcement. With well over 2 million pools between the two states, roughly 40% of the U.S. pool industry is now governed by at least some energy efficiency requirements, with more states certain to follow.

In many communities, utilities are adopting rebate programs to encourage pool owners to replace standard pumps with energy-efficient models, such as variable-speed pumps. Historically, most pool pumps have used single-speed motors, often with undersized plumbing and filters that strain the pump even further. But two-speed or multispeed pumps, which provide some options for greater efficiency, are becoming increasingly common; and so are variable-speed pumps, which can be up to 90% more efficient than a single-speed motor.


2.

Here's how it works: In a typical one- or two-speed pump, motor speed is almost always locked at too high a flow rate, wasting energy. In many mechanical applications, the lowest effective speed will be the most efficient, whether that is for propelling a vehicle or powering various combinations of pool features.

The best variable-speed pumps can be programmed to meet the requirements of each task that they must perform—anywhere from 400 rpm for the filter alone to 3,450 rpm for suction cleaners, water features, spa and whatever other applications may be included on a specific pool. Some models feature a similar rpm range, but are locked to a half-dozen or so settings. This makes them less flexible than the best variable-speed pumps, but still much more efficient than single-speed models.

Because of this new technology, pool professionals have had to change the way they view pump power. The horsepower measure just doesn't apply. Whereas a standard pump weighs in at 1 or 2 horse, a variable-speed pump weighs in at 0 to 2 or 3 horse, depending on the manufacturer.

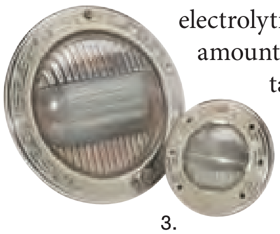
Table 1. Top Annual Average Electric Usage (kWh) per Home Appliance

APPLIANCE	PG&E	SDG&E	SCE	LADWP
Pool pump	2,580	2,557	2,772	3,096
Spa electric heater	1,346	903	2,514	895
Central air	1,108	644	1,494	1,075
Refrigerator	788	780	801	754

1. The IntelliTouch ScreenLogic PDA
2. The Intelliflo Pump
3. Intellibrite Lights

(Granted, the 0 is only technically accurate if the pump isn't running; the low end is really a small fraction of a horse above 0.) When paired with automation software, a variable-speed pump can automatically respond with additional power for cleaning, spa operations, waterfalls, or fountains, and can dial back the power for normal filtration. It doesn't matter whether the job calls for 1 horse or 3, because the pump is programmed to adjust automatically.

The slower speeds are not only more energy efficient, but quieter. They also reduce wear on equipment, such as filters and chlorinators.


3.

Switching to a variable-speed pump can save pool owners in California over $1,000 per year. Savings have also been significant in states where electricity is cheaper.

Control Systems

Digital automation systems allow pool owners to schedule cleaning and filtration, and to control the temperature for pools and spas. Similar to automation systems found inside the home, pool control systems ensure that equipment will never be left running accidentally or be operated at a higher level than necessary. Homeowners can also program these systems to take advantage of off-peak or seasonal utility rates or to optimize functions during times of heavy or light use. Interface options include wireless remotes, wall-mounted LCD touch screens, and PDAs.

Heaters

When Al Gore renovated his Nashville home to meet the highest of green building standards, one of the areas of top concern was his pool. In addition to many of the technology upgrades available, Gore utilized a geothermal heating system that was able to conduct heat from the ground into the water pipes to heat his pool.

Solar-heating systems are also available, ranging from rudimentary pool covers to large solar panels. This technology is an ideal option for many homes, but like any solar technology, it can be prohibitively expensive for the average homeowner, and it is not appropriate in all climates.

Salt-Chlorine Systems

An automatic chlorine generator is another popular equipment option that utilizes onboard electronics and automation technology. A chlorine generator converts ordinary table salt—roughly 1 teaspoon per gallon—into sodium and chlorine through a process known as electrolytic chlorine generation. This amount of salt is beneath the human taste threshold, and allows the homeowner to forego standard chlorine additives.

The lower flow rates of a variable-speed pump are compatible with a chlorine generator, as it is difficult to move water for skimming purposes without creating enough motion to generate chlorine. The minimum flow rate for every system I am familiar with is around 20 gallons per minute, which is a very low flow rate for filtration purposes. The most common recommended low speed for variable-speed pumps is 30 gallons per minute.

LED Underwater Lights

LED pool lights are much more energy efficient than traditional incandescent and halogen pool lights. Pentair's LED pool lights use only 37 watts to create the same light output as its halogen pool light does at 161 watts. (And the LEDs change color, too!) LEDs are also more durable and last longer than traditional incandescent bulbs. The estimated life of a LED pool light is 30,000– to 50,000 hours, compared to only 5,000 hours for a traditional incandescent bulb.

A Green Pool

When seeking out ways to design and build homes for efficiency and a smaller environmental footprint, the backyard still probably isn't going to be the top concern for most homeowners, but it definitely shouldn't be the last.

Whether new construction or a renovation, home owners with swimming pools are finding that the energy efficiency impact of some relatively simple pool equipment upgrades rival that of efficiency-minded features inside the home. And when considering how comparatively inefficient the standard pool equipment can be, the pool becomes an issue that no green-minded homeowner will want to ignore.

—Jeff Farlow

Jeff Farlow is program manager for Energy Initiatives at Pentair Water Pool and Spa.

EFFICIENCY OPTIONS & OPPORTUNITIES

In our Energy Efficient Commercial Pools program (EECPP), we have performed over 150 audits of commercial and public swimming pools and spas. What we found is a formula for reducing pool-related energy use by 60%–80%, on average. This formula also applies directly to private residential pools. The key to saving energy on pools, spas, ponds, fountains, and so on is to reduce the system head pressure. When this is done, the pump motor works less and so uses less energy to do its job.

How do you reduce the system head? Simply do any or all of the following things:

- Slow the water down (using a variable-speed pump).
- Increase the pipe diameter.
- Straighten the bends in the pool piping (sweep or flex elbows; design the system in a straight line).
- Remove miscellaneous restrictions (such as valves and reducers).
- Use large-cartridge filters.
- Bypass heaters when they are not in use.

My experience with swimming pools is that in the pool service industry, a sector not known for adopting readily to change, forward-thinking "pool guys" are making a lot of money selling energy-efficient retrofits to their customers. California pool owners are paying over $100 per month in electric costs just to filter their pools. Knowledgeable pool professionals can bring that cost down to $20–$30 per month, which will typically pay off the investment in two to three years.

—Scott Clay

Scott Clay is a program manager at the Pacific Gas and Electric Company

For more information:

Pentair Water Pool and Spa manufactures and sells versions of all of the pool technology described in the article. To find out more, go to www.pentairpool.com.